

ANESS RAHMANI

Full-Stack Developer | Next.js | FastAPI | React | Angular | Node.js | PostgreSQL | MongoDB

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Full-stack developer building both ends of the stack: Next.js/React frontends and FastAPI backends with PostgreSQL, Redis, and Celery. Delivered a full e-commerce platform end-to-end (auth, caching, async task processing, containerized deployment) plus a standalone auth service and a real-time monitoring platform. Comfortable owning a feature from database schema to deployed UI. Trilingual: Arabic (native), English (C2), French (B2-C1).

TECHNICAL SKILLS

Frontend: React, Next.js (App Router, Server Components, SSR/SSG/ISR), TypeScript, Tailwind CSS, shadcn/ui, TanStack Query, React Hook Form + Zod

Backend: FastAPI, Python, Node.js, Express.js, RESTful API design, Pydantic, async/await

Databases: PostgreSQL, SQLAlchemy/SQLModel, Alembic migrations, indexing/query optimization, MongoDB

Caching & Messaging: Redis (cache-aside, rate limiting), Celery (async task queues, retries)

Auth & Security: JWT (access/refresh rotation), OAuth2/OIDC (Google sign-in), bcrypt, RBAC/ABAC, CORS

DevOps & Testing: Docker, Docker Compose, multi-stage builds, Pytest, Jest, React Testing Library

Tools & Deployment: Git, GitHub, Vercel, Render, Postman

Other: PyTorch (applied ML/CV background)

Languages: Arabic (native), English (C2), French (B2-C1)

PROJECTS

E-Commerce Platform | *Next.js · FastAPI · PostgreSQL · Redis · Celery · Docker* | 2026

- Built the full stack: Next.js App Router storefront/checkout (server components, TanStack Query, React Hook Form + Zod) on a FastAPI backend with a relational PostgreSQL schema (SQLAlchemy/SQLModel, Alembic migrations)
- Integrated JWT/Google OAuth2 auth end-to-end, with protected routes on the frontend and RBAC-gated endpoints on the backend
- Added Redis cache-aside for product listings and rate limiting on auth routes; offloaded checkout/order processing to Celery background workers
- Containerized the entire stack with Docker Compose (API, PostgreSQL, Redis, Celery worker); wrote Pytest coverage for auth/API and Jest/RTL coverage for key UI flows
- Deployed the frontend on Vercel and backend on Render, wiring cross-origin auth and environment config between them

SecureAuth API | *FastAPI · JWT · OAuth2 · Redis* | 2026

- Built a standalone, reusable authentication service: JWT access/refresh flow, Google OAuth2/OIDC, bcrypt password hashing, Redis-backed sessions, and RBAC/ABAC middleware
- Added rate limiting and Pydantic validation; Dockerized for drop-in use across other backend projects

AcciSense — Real-Time Accident Detection Platform | *FastAPI · React · MQTT · WebSocket · Supabase* | 2025 – 2026

- Master's thesis project (defended at 19.5/20): benchmarked three video-understanding architectures (R3D-18, SlowFast R50, Video Swin Transformer) for road accident detection and severity estimation, reaching 87.3% detection accuracy
- Built the full production platform: FastAPI backend, React/Vite frontend with real-time WebSocket updates, MQTT monitoring pipeline, and Supabase logging

Harbor Fish Log — Client Delivery | *React · FastAPI · MongoDB* | 2025

- Delivered a mobile-optimized field data logging app for a PhD researcher, replacing manual spreadsheet workflows; records log in under 30 seconds with instant search, filter, and CSV/JSON export

IoT Device Monitoring Dashboard | *FastAPI · React · MQTT* | 2023

- Real-time dashboard for IoT device monitoring: FastAPI backend, React frontend, MQTT subscriber for continuous device data streaming and live visualization

EXPERIENCE

Web Developer | Freelance Client | 2025 | Remote

- Designed, built, and delivered Harbor Fish Log — a full-stack web application — for a doctoral researcher, from requirements gathering to deployment

Data Annotation Specialist | **Freelance (Upwork / LinkedIn)** | 2022 – 2024 | Remote

- Completed data cleaning, annotation, and proofreading contracts for research and commercial clients

IT Department Head | **University Club** | 2021 – 2023 | Khemis Miliana

- Led technical initiatives and managed digital infrastructure for the organization

EDUCATION

Master's Degree in Telecommunication Systems | **Université Djilali Bounaama** | 2024 – 2026 | Algeria

- Thesis: Automatic Road Accident Detection and Severity Estimation Using Deep Learning Models — 19.5/20

Bachelor's Degree in Electrical Engineering | **Université Djilali Bounaama** | 2020 – 2023 | Algeria